BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 148 · Integrated Development Environments

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does an interpreter normally do?

Q2. Q = (A AND B) OR C. A=1, B=0, C=1. What is Q?
A. 0
B. 2
C. 1
D. 10

Q3. “A count-controlled loop is only suitable when the number of repetitions can never be known.” Correct this. Explain why.

Q4. A program keeps asking for a password until its stopping condition is met. “A condition-controlled loop always runs ten times.” Correct this. What controls when it stops?

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I Do · Read new knowledge

Key vocabulary

Editor — Provides tools for entering and organising source code.

Syntax highlighting and completion — Make code structure and possible mistakes easier to notice.

Breakpoints and debugging — Pause execution to inspect variables and follow faulty logic.

Runtime/output tools — Allow programmers to run/test code and view results from the environment.

New learning

Editor: Provides tools for entering and organising source code.

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Runtime/output tools: Allow programmers to run/test code and view results from the environment.

Worked example

A program produces 9 instead of expected 10. Explain how a debugger could locate the error.

1. Set a breakpoint before the suspect calculation.

2. Run to the breakpoint.

3. Inspect current variable values and step line by line.

4. Find first unexpected update, fix it and rerun tests.

We Do · Practise with guidance

Guided 1

A pupil has misspelled a variable. Identify two IDE features that may help.

Worked method: Test after correction.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the faulty bits-to-bytes calculation and check its output.

```
bits = 40
bytes = bits * 8
print(bytes)
```

iterate

Question 1

Name two IDE features. [2 marks]

Question 2

Explain why a breakpoint helps a developer. [3 marks]

Question 3

State why a program should be tested after fixing an IDE-reported syntax error. [2 marks]

Exit Ticket · Answer before checking

A breakpoint changes all incorrect variables automatically.

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Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. An interpreter translates and runs code as it goes, often one statement at a time. This gives quick feedback during testing.

Q2. C — 1

Q3. When the number of repetitions is known or determined by a fixed count.

Q4. A condition-controlled loop repeats until a condition tells it to stop. It need not run ten times. Choose the loop to suit the task and trace its steps.

I Do / We Do pages · Guided practice

Q1. Test after correction.

You Do pages · Code First

```
bits = 40
bytes = bits DIV 8
print(bytes)
```

Expected result: 5

You Do pages · Independent questions

Q1. Code editor, syntax highlighting, autocomplete, debugging, breakpoints, error diagnostics or run console.

Q2. Pauses execution at a selected point, allowing variables/program state to be inspected.

Q3. Syntax correction only ensures code can be parsed; logical behaviour still needs checking.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: